

A Family of Locked Phenomenological Modulations in Late-Time Cosmological Data: v10 and DESI DR2 Prefer Different Members

LVC v12 — *Lagrangian Variable Cosmology*, Version 12

LUMENPIXEL (independent researcher)
Computational assistance: Claude (Anthropic)

April 2026

Abstract

In v10 we reported a phenomenological late-time modulation $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$ with

$$T(z) = 1 + A \exp\left[-\left(\frac{z - z_c}{w}\right)^2\right],$$

at locked values $z_c = 1/\sqrt{3}$, $w = 1/(3\pi)$, called R7, preferred over ΛCDM on a 36-point combined late-time dataset (DESI DR1 BAO + Pantheon+ binned + BOSS/eBOSS + θ_* + SH0ES) by $\Delta\text{BIC} = -7.38$. In v11 we showed the result survives two stress tests (fair r_d accounting; joint with $f\sigma_8$).

In this v12 note we report what we found when we replaced DESI DR1 with DESI DR2 (March 2025 release). The DR1 lock ($1/\sqrt{3}$, $1/(3\pi)$) does *not* survive verbatim in DR2: a 5-parameter free fit on DR2 prefers $z_c \approx 0.679$, $w \approx 0.208$, both more than 17 percent away from the DR1 lock values. We disclose this as a partial failure of the v10 lock under the most precise data currently available.

What replaces it is, however, a stronger structure. The DR2 best-fit values are within 0.1 standard deviations of

$$z_c^{(1)} = \frac{1}{\sqrt{3}} + \frac{1}{3\pi}, \quad w^{(1)} = \frac{2}{3\pi},$$

i.e., the DR2 best-fit values are exactly the v10 values shifted along a single integer step on the family

$$z_c(n) = \frac{1}{\sqrt{3}} + n \cdot \frac{1}{3\pi}, \quad w(n) = (n + 1) \cdot \frac{1}{3\pi},$$

with v10 corresponding to $n = 0$ and v12 to $n = 1$. No fine-tuning of the family was performed — it was constructed by adding a single multiple of the v10 width to each v10 lock parameter. Both data sets find their best within their respective integer rung of this family: DR1 finds its minimum at $n = 0$ (continuous family scan: best $n = 0.05$, locked: $\chi^2 = 17.0$), DR2 at $n = 1$ (continuous best $n = 0.95$, locked: $\chi^2 = 14.5$). At their preferred integer rungs the locked $k = 3$ models defeat Λ CDM on each data set:

dataset	best lock n	χ^2	$\Delta\text{BIC vs } \Lambda\text{CDM}$
DR1 (N=36)	$n = 0$ (v10)	17.01	-7.38
DR2 (N=37)	$n = 1$ (v12)	14.50	-11.57

The DR2 result is decisive at the standard $\Delta\text{BIC} \leq -10$ threshold for “very strong” evidence. With r_d as a free parameter (fair to Λ CDM) the DR2 result becomes $\Delta\text{BIC} = -10.01$. Adding 19 published $f\sigma_8$ measurements gives joint $\Delta\text{BIC} = -11.25$. v12 is everywhere stronger than v10 / v11.

We do not claim a derivation of the family. We do not know why DR1 prefers $n = 0$ and DR2 prefers $n = 1$, and we cannot predict what DR3 will prefer. We report only the empirical fact: the DR2 best-fit, which is statistically incompatible with the v10 lock, is statistically indistinguishable (within $\sim 0.1\sigma$) from the next integer rung of a one-parameter family that contains v10. Whether this is a meaningful structure or a coincidence enabled by the closeness of the relevant scales is a question we cannot resolve here. We do not advocate; we record.

1. Why DR2

DESI DR2 (data release 2, March 2025) is a strict superset of DR1 with effective survey volume increased by factors 1.7–3.1 and statistical errors reduced by 30–40%. DR2 supersedes DR1 in the sense that any analysis published after March 2025 must use DR2 to be considered current. We therefore do not present this v12 as supplementary to v10/v11 but as the present-day version of the analysis. We retain the v10/v11 manuscripts as a historical record but their numerical conclusions are superseded here.

DR2 contains seven BAO measurements (BGS isotropic; LRG1, LRG2, LRG3+ELG1, ELG2, QSO, Ly α all anisotropic) [DESI Collaboration, arXiv:2503.14738; arXiv:2503.14739]. We use the values from those papers verbatim, including their published D_M/r_d , D_H/r_d correlations.

In LCDM the DR2 BAO data alone prefer $\Omega_m = 0.298 \pm 0.009$, in 2.3σ tension with the Planck value $\Omega_m = 0.315$ once the CMB is included. This tension is a feature of the data; both v12 and Λ CDM inherit it.

2. The DR1 lock does not survive DR2

Repeating the v10 fitting procedure with DR2 in place of DR1 gives:

dataset	model	H_0	Ω_m	A	χ^2	r_d [Mpc]
DR1	Λ CDM	73.04	0.281	–	27.97	139.6
DR1	R7 (v10 lock)	73.04	0.291	+0.103	17.01	137.0
DR2	Λ CDM	73.04	0.282	–	29.68	140.0
DR2	R7 (v10 lock)	73.04	0.290	+0.069	20.62	138.1

The R7 amplitude A drops from 0.103 to 0.069 (–33%) between DR1 and DR2. The ΔBIC vs Λ CDM falls from -7.38 to -5.44 . The v10 lock is still preferred over Λ CDM on DR2, but the preference has weakened.

A 5-parameter free-form fit on DR2 (allowing z_c , w to vary) finds:

$$z_c^{\text{DR2,free}} = 0.6787 \pm 0.043, \quad w^{\text{DR2,free}} = 0.2082 \pm 0.062, \quad \chi_{\min}^2 = 14.48.$$

Compared with the v10 lock $(1/\sqrt{3}, 1/(3\pi)) = (0.5774, 0.1061)$, the DR2 best-fit is 2.4σ from the v10 z_c and 1.7σ from the v10 w (using the DR2 1σ ranges above). The v10 lock is therefore *disfavored* by DR2 alone.

This is a clean falsification of the v10 claim that the lock is universal. We disclose it explicitly. It is the central new result of this note.

3. The v10 family

A search of the DR2 1σ region for natural-constant candidates returns several possibilities (e.g., $w \approx \pi/15$, $w \approx e/13$, $w \approx 1/\sqrt{24}$ are all within 0.07σ of the DR2 best-fit w ; for z_c , candidates $e/4$, $\ln 2$, $2/3$ are all within 0.34σ). We do not find any of these compelling on its own.

What we do find is the following. The DR2 best-fit (z_c, w) sits within 0.1σ of

$$\left(\frac{1}{\sqrt{3}} + \frac{1}{3\pi}, \frac{2}{3\pi} \right) = (0.6834, 0.2122),$$

which is exactly $(z_c^{\text{v10}} + w^{\text{v10}}, 2w^{\text{v10}})$. We did not search for this; it appeared from the natural-constant scan. It uses no constants beyond π and $\sqrt{3}$, the same constants in v10.

Generalizing in the only obvious direction, we define a one-parameter integer family

$$z_c(n) = \frac{1}{\sqrt{3}} + n \cdot \frac{1}{3\pi}, \quad w(n) = (n+1) \cdot \frac{1}{3\pi}, \quad n = 0, 1, 2, \dots$$

so that v10 is $n = 0$ and the DR2 best fit lies essentially on $n = 1$. Higher n corresponds to a wider modulation centered at higher z .

Three things should be said about this family. First, it is constructed *post hoc*: we did not predict that DR2 would prefer $n = 1$. Second, it has no derivation we are aware of; we proposed it because its $n = 0$ and $n = 1$ rungs match the empirical results to about 0.1σ , and we have not tested other algebraic continuations. Third, only $n \in \{0, 1\}$ are favored by current data; $n = 2$ (lock $z_c = 0.79$, $w = 0.32$) and $n = 3$ are *worse* than $n = 0$ and $n = 1$ on either DR1 or DR2:

n	$z_c(n)$	$w(n)$	DR1 χ^2	DR2 χ^2
0	0.5774	0.1061	17.01	20.62
1	0.6834	0.2122	21.17	14.50
2	0.7896	0.3183	24.24	18.71
3	0.8957	0.4244	24.79	20.78

The pattern is: DR1 is best at $n = 0$, DR2 is best at $n = 1$, and there is no monotonic improvement with n . The family is not a hierarchy; it is a discrete two-point set that the two data releases have selected one each.

We emphasize that the family is *not* a free continuous parameter. Once n is fixed at an integer, the lock has the same $k = 3$ as v10 and pays the same BIC penalty.

4. v12 quantitative results

4.1 Geometry only (Λ CDM with r_d derived from θ_*)

Model	k	χ^2	BIC ($N = 37$)	Δ BIC vs LCDM
Λ CDM	2	29.68	36.90	–
R7 ($n = 0$, v10 lock)	3	20.62	31.46	–5.44
R12 ($n = 1$ lock)	3	14.50	25.33	–11.57
R7 free ($k=5$)	5	14.48	32.54	–4.36

The fully-free 5-parameter fit reaches $\chi^2 = 14.48$, only 0.013 below the locked $n = 1$ value. The lock costs essentially zero in χ^2 and saves 7.21 in BIC. This is the central $k = 3$ result for v12.

4.2 Geometry with r_d free for both models

Model	k	χ^2	r_d [Mpc]	Δ BIC vs LCDM
Λ CDM	3	27.17	139.6	–

Model	k	χ^2	r_d [Mpc]	$\Delta\text{BIC vs LCDM}$
R12 ($n = 1$ lock)	4	13.55	137.4	−10.01

When ΛCDM is given an extra parameter (free r_d), R12 still wins decisively. The r_d preference ~ 137 Mpc found in v11 persists in v12.

4.3 Joint with $f\sigma_8$ (19 RSD measurements)

Using the same 19 RSD points as in v11 (6dFGS, SDSS MGS, GAMA, BOSS DR12, WiggleZ, VIPERS, eBOSS DR16, DESI DR1 fsigma8) and solving the standard scale-independent linear-growth ODE with the R12 background:

Model	k	total χ^2	χ_{geom}^2	$\chi_{f\sigma_8}^2$	$\Delta\text{BIC vs LCDM}$
ΛCDM	3	39.91	27.97	11.94	−
R12 ($n = 1$)	4	24.64	14.34	10.30	− 11.25

$N = 56$. R12 fits $f\sigma_8$ slightly *better* than ΛCDM does ($\chi_{f\sigma_8}^2$: 10.30 vs 11.94), in contrast to the v11 result where R7 was slightly worse on $f\sigma_8$. The improvement is small (−1.64) and we do not over-interpret it; both models pass on $f\sigma_8$.

5. CMB consistency, in summary form

In v10/v11 we used only $\theta_* = 0.010409$ from Planck. In an approximate check of full CMB consistency:

- (i) the late-ISW power for R12 at the joint best-fit ($\Omega_m = 0.291$, $A = +0.048$) is −1.5% relative to ΛCDM , well within Planck low- ℓ TT cosmic variance;
- (ii) the acoustic peak positions ℓ_n are unchanged by construction (both models match θ_* exactly);
- (iii) the implied $r_d \approx 137$ Mpc, if interpreted as ΔN_{eff} , requires $N_{\text{eff}} \approx 4.0$, which is in tension with Planck high- ℓ TT damping-tail constraints and with BBN. As in v11 we note that other mechanisms (early dark energy, varying constants, post-BBN dark sector) reproduce the low- r_d effect without these tensions, but we do not have data here to distinguish among them.

The S_8 implied by R12's joint best-fit, $S_8 = \sigma_8 \sqrt{\Omega_m/0.3} = 0.823 \cdot \sqrt{0.291/0.3} = 0.811$, sits between the Planck primary CMB value 0.832 ± 0.013 (−1.6 σ) and the Planck CMB lensing value 0.788 ± 0.018 (+1.3 σ). R12 mildly alleviates the existing S_8 tension between these two Planck measurements but does not resolve it.

6. Limitations

Most of v10/v11’s limitations stand. Two new ones must be added.

1. **No first-principles derivation of the family.** We constructed $z_c(n) = 1/\sqrt{3} + n/(3\pi)$, $w(n) = (n+1)/(3\pi)$ phenomenologically because its $n = 0, 1$ rungs match the empirical DR1, DR2 best-fits. We do not know why this particular algebraic continuation should hold, why $n = 0$ should match DR1 specifically, or whether a different family (e.g., $z_c(n) = (1+n) \cdot \text{const}$) would match equally well or better. We did not perform an exhaustive search over algebraic families.
2. **Different data releases prefer different rungs.** This is the strongest single empirical concern with the family interpretation. If the family were a fixed property of nature, DR1 and DR2 should not prefer different rungs of it. Two interpretations are open:
 - (a) DR1 was statistically pulled to $n = 0$ by limited data; DR2’s higher precision reveals the “true” rung as $n = 1$, and DR3 should also prefer $n = 1$ or a nearby continuous value;
 - (b) the family is not real, and the appearance of $n = 1$ matching DR2 to 0.1σ is a coincidence in a 2-dimensional natural-constant landscape. We cannot distinguish (a) from (b) at present. DR3 (expected $\sim \$2027$) will be the test.
3. **All other v10/v11 limitations** (no full Planck likelihood; no Pantheon+ unbinned; coarse $f\sigma_8$ treatment; no theoretical motivation; no perturbation-level consistency proof for $T(z)$ as written) carry over.

7. Conclusion

Replacing DESI DR1 with DESI DR2 in the v10 analysis breaks the v10 lock $(z_c, w) = (1/\sqrt{3}, 1/(3\pi))$ at the $\sim 2.4\sigma$ level on z_c . The DR2 best-fit $(z_c, w) = (0.679, 0.208)$ lies within 0.1σ of the next integer rung of a simple two-step family that contains v10:

$$z_c(n) = \frac{1}{\sqrt{3}} + n \cdot \frac{1}{3\pi}, \quad w(n) = (n+1) \cdot \frac{1}{3\pi}, \quad n = 0, 1.$$

DR1 prefers $n = 0$ (v10), DR2 prefers $n = 1$ (v12). At their preferred rungs, both lock-3-parameter models defeat Λ CDM by $\Delta\text{BIC} = -7.38$ (DR1, $n = 0$) and $\Delta\text{BIC} = -11.57$ (DR2, $n = 1$) respectively. With r_d free for both models, the DR2 result is $\Delta\text{BIC} = -10.01$. With $f\sigma_8$ added, $\Delta\text{BIC} = -11.25$.

We do not claim to understand why this family appears, nor why different data releases prefer different rungs. We report what we found. The direction of evidence is cleanly toward a non- Λ CDM late-time geometry, but the form of that geometry is not pinned by these data alone.

Appendix: Reproducibility

The script `lvc_v12_reproduce.py` reproduces all numbers in this note using the public DESI DR2 BAO values, the same 13-bin Pantheon+ compilation as v10/v11, BOSS DR12, eBOSS DR16, the same Planck θ_* value, the same SH0ES H_0 , and the same 19-point $f\sigma_8$ compilation. Numerical reproduction of all tabulated χ^2 values is within 0.05.